

Customer Churn Prediction Using Logistic Regression

Project Overview

The goal of the project was to predict customer churn for a telecommunications company by developing a machine learning model capable of doing so. Customer churn occurs when a customer discontinues service, resulting in lost revenue and increased customer acquisition costs. By identifying customers at risk of churning, businesses can implement targeted retention strategies and improve long-term profitability.

Dataset

This dataset initially contained 7,043 customer records and 21 variables describing customer demographics, account information, services subscribed to, billing details, and churn status.

The target Variable was “Churn” : Yes = customer churned, No = customer remained with the company

After data cleaning, the final dataset contained 7,032 customer records.

Data Preparation

Several processing steps were performed before model training:

1. Loaded and inspected the dataset
2. Examines data types and missing values
3. Converted the “TotalCharges” column from string format to numeric format
4. Identified 11 hidden missing values during conversion
5. Removed records containing missing values
6. Converted the target variable, “Churn”
 - Yes = 1
 - No = 0
7. Removed the “customerID” column because it provided no predictive value
8. Applied one hot encoding to categorical variables using `pd.get_dummies()`
9. Split the dataset into training and testing sets using an 80/20 split
10. Standardized the feature values using “StandardScaler”

Exploratory Data Analysis

Churn Distribution

The dataset contained:

- 5,163 retained customers (73.4%)
- 1, 869 churned customers (26.6%)

The dataset was moderately imbalanced but still suitable for classification modeling

Contract Type Analysis

Contract length showed a strong relationship with churn behavior

Contract Type	Churn Rate
Month to month	42.7%
One year	11.3%
Two Years	2.8%

This supports the idea that customers with longer term contracts are noticeably less likely to leave the company.

Model Development

A Logistic Regression classifier was selected because:

- The target variable is binary
- Logistic Regression is interpretable
- Coefficients can be analyzed to understand feature importance

The model was trained using the processed training dataset and evaluated on previously unseen test data

Model Performance

Metric	Score
Accuracy	80.4%
Precision	64.8%
Recall	57.5%
ROC-AUC	83.6%

Interpretation:

Accuracy (80.4%)

- The model correctly classified 80% of customers

Precision (64.8%)

- When the model predicted churn, it was correct approx. 65% of the time

Recall (57.5%)

- The model successfully identified approx. 58% of customers who actually churned
ROC-AUC (83.6%)
- The model demonstrated good ability to distinguish between churning and non-churning customers

The model significantly outperformed a baseline approach of predicting that all customers would remain with the company.

Confusion Matrix

```
[[916 117]
 [159 215]]
```

Interpretation:

- 916 customers were correctly predicted to stay
- 215 customers were correctly predicted to churn
- 117 customers were incorrectly predicted to churn
- 159 customers were incorrectly predicted to stay

The most important errors were false negatives because these represent customers who churned without being identified by the model

Feature Importance Analysis

Features Associated with Increased Churn Risk:

- Fiber Optic Internet Service
- Higher Total Charges
- Streaming TV Service
- Streaming Movie Service
- Multiple Lines
- Electronic Check Payment Method

Features Associated with Reduced Churn Risk

- Longer Customer Tenure
- Two-Year Contracts
- One-Year Contracts
- Online Security Service
- Technical Support Service

The strongest retention factor identified by the model was customer tenure, while the strongest churn associated factor was optic internet service.

Business Recommendations

Based on the analysis, the company should consider:

1. Developing retention programs targeted at month-to-month customers.

2. Encouraging customers to move to one-year or two-year contracts.
3. Investigating the higher churn rates among fiber optic internet customers.
4. Promoting online security and technical support services.
5. Identifying high-risk electronic-check customers for proactive outreach.

Conclusion

This project successfully developed a customer churn prediction model using Logistic Regression. The model achieved strong overall performance with an ROC-AUC score of 83.6% and identified several important factors associated with customer retention and churn. The findings provide actionable business insights that could help reduce customer attrition and improve long-term customer retention strategies.